

3(a)(i)	A – G1 / gap 1 ; B – S / synthesis ; A DNA replication I S1 / S2 / replication unqualified	2																
3(a)(ii)	telophase ;	1																
3(a)(iii)	<i>any two from:</i> increase in cytoplasm / increase in number of (named) organelle(s), <u>during interphase</u> ; A synthesis of organelles I cell growth (re)formation of <u>nuclear envelope</u> ; nuclear envelope, forms around each group of chromosomes ; A nuclear membrane as an ECF if used for mp2 (movement of) organelles to be shared between two daughter cells ; spindle, disassembles / breaks down / degrades / disappears / AW ; cleavage furrow forms ; A cytoplasm / cell membrane, pinches in / constricts / infolds AVP ; e.g. (actin) microfilaments / microtubules, form contractile ring / AW, around equator of cell / beneath cell surface membrane A cytoskeleton for microtubules / microfilaments	2																
3(a)(iv)	<i>one mark per row</i> <table><tr><td></td><td colspan="3">stages of cell cycle</td></tr><tr><td></td><td>A</td><td>B</td><td>D</td></tr><tr><td>number of nuclei within the stem cell</td><td>1</td><td>1</td><td>2 ;</td></tr><tr><td>number of chromosomes ineach nucleus</td><td>12</td><td>12</td><td>12 ;</td></tr></table>		stages of cell cycle				A	B	D	number of nuclei within the stem cell	1	1	2 ;	number of chromosomes ineach nucleus	12	12	12 ;	2
	stages of cell cycle																	
	A	B	D															
number of nuclei within the stem cell	1	1	2 ;															
number of chromosomes ineach nucleus	12	12	12 ;															

Question	Answer	Marks
3(b)	1 stem cells) continue to divide / divide repeatedly (by mitosis) ; A at least one daughter cell, retains ability to divide / is a stem cell / A 'repeated mitosis' / A 'to produce more stem cells' <i>any two from mps 2-5: to produce cells</i> 2 for growth ; 3 for repair of tissues ; R repair of cells 4 for replacement of, worn out / old / dead, cells ; A damaged cells if repair of tissues not given 5 to differentiate for formation of, tissues / organs ; A specialised cells for forming, tissue / organs <i>and:</i> 6 AVP ; e.g. <i>ref. to</i> potency of stem cells (multipotent or pluripotent)	3